Amit Kumar Behera

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Research Focus: Applied AI/ML · Edge AI Deployment · Resource-Constrained Inference · Computer Vision · Secure IoT Systems

Degree	Institute	Year	CPI
BS – Computer Science	Indian Institute of Technology, Patna	2024-2028	**9.51**
Senior Secondary (XII)	Kendriya Vidyalaya DRDO, Hyderabad	2023	8.1
Secondary (X)	Kendriya Vidyalaya NAD, Visakhapatnam	2021	**10.0**

Selected Publications

- **Edge-Based Multi-Environment LoRa Communication for Sustainable IoT Deployments** – *IEEE INSTCon 2026.* **1st Author, Best Paper Award**
- **Performance–Efficiency Tradeoff of Deep CNN Models for Brain Tumor MRI Classification on Edge Devices** – *IEEE DICCT 2026.* **1st Author, Best Paper Award**
- **EdgeGuard: Lightweight AI-Driven IDS for Real-Time Cyberattack Mitigation on Edge Devices** – *IEEE AIML Applications Conference.* **2nd Author, Best Paper Award**
- **EdgeISA: A Knowledge Distillation-Based Lightweight Framework for Instruction Set Architecture Detection in Firmware Analysis** – *IEEE UPWIECON 2026.* **1st Author, Accepted**
- **Learning Earth Observation Embeddings with Temporal & Non-Temporal Self-Supervision using RCF** – *Springer LNNS, ITAI 2026.* **2nd Author**

Under Review: *AI-Driven Blood Group Detection* (**1st Author**, IEEE ICAIHC 2026); *AI-Based Condition-Based Maintenance of Radar Antenna Motors* (**2nd Author**). **Full publication list:** Google Scholar

Research Experience

Research Intern, Scientific Analysis Group (SAG), DRDO *Jun. 2026 – Jul. 2026*
Trusted Computing & Edge AI for Secure Embedded Platforms

- Investigated **TPM, secure boot, and boot-sequence verification**; developed a lightweight **ISA detection pipeline** for resource-constrained embedded systems, with work accepted at **IEEE UPWIECON 2026**.

Research Collaborator, Indian Air Force *Feb. 2026 – Present*
AI-Driven Condition Monitoring of Radar Antenna Systems

- Analyzed **437,000+ vibration and acoustic measurements** from operational radar systems; developed ML-based fault-diagnosis pipelines using **FFT, PSD, STFT, spectrograms, and feature engineering**.

Research Intern, NESAC, Department of Space, Government of India *Dec. 2025 – Jan. 2026*
Self-Supervised Learning for Earth Observation

- Developed an **RCF-based self-supervised embedding pipeline** spanning **300K+ geographic locations**; benchmarked temporal satellite representations, with work accepted at **Springer LNNS, ITAI 2026**.

Research Intern, NIT Patna *Oct. 2025 – Dec. 2025*
Multilingual LLMs for Humor Generation

- Developed and evaluated a multilingual LLM pipeline on the **MWAHAHA dataset**, combining prompt engineering, constraint handling, and response reranking.

Technical Skills

Languages	Python, C/C++, SQL, Linux
AI / ML	PyTorch, Transformers, Scikit-learn, XGBoost, DINO/DINOv2
Computer Vision	OpenCV, YOLOv5, RT-DETR, Image Processing
Edge AI	ONNX, TensorRT, INT8 Quantization, Knowledge Distillation, Raspberry Pi, Jetson Nano
Research	Self-Supervised Learning, Attention Mechanisms, SHAP, Feature Selection

Projects

DINO-EFS: Entropy-Guided Edge Intelligence Framework *2025 – Present*
Lightweight self-supervised visual learning for edge deployment; IIT Patna.

- Developed a visual classification pipeline combining **DINO/DINOv2, entropy-guided attention, E-CGLEO feature selection, and prototype-based classification**, achieving **99.76%, 93.19%, and 96.17% accuracy** across three benchmarks.
- Reduced feature dimensionality by **60–70%** with **<1% accuracy drop**; achieved **∼20× compression** and **13.17 ms inference at 75.9 FPS** using FP16 on Raspberry Pi 5.

Edge-IoT Energy Forecasting System *2025 – Present*
Explainable residential energy forecasting with edge-IoT deployment; IIT Patna & NIELIT Patna.

- Built a **Raspberry Pi 5 and LoRa SX1262** sensing pipeline with CRC validation and adaptive duty cycling, reducing **cloud bandwidth by 41%**.
- Evaluated **FT-Transformer, TabPFN, and TabNet**, achieving $R^2 = 0.9747$; deployed an ONNX-quantized model with **14.2 ms inference** and SHAP-based explanations.

EdgeGuard: AI-Powered Intrusion Detection System *2025 – Present*
Real-time network threat detection and automated traffic enforcement on edge devices.

- Developed a Raspberry Pi-based intrusion detection system achieving **98% detection with <2% false positives** on known threats through ML-based traffic analysis and automated enforcement.
- Integrated **Random Forest, XGBoost, iptables, and Flask** for threat detection and traffic control; secured **1st place among 1,000+ teams**, earning an **INR 2.5 lakh prize** and startup support.

Awards & Achievements

- **Hackathon Winner** — 5 wins across 5 hackathons, including 4 national-level competitions
- **Aignite Cybersecurity Hackathon** — 1st Place among 1,000+ teams; INR 2.5L prize (2025)
- **Samsung Grade-I Fellowship** — IISc Bengaluru; 100% fee waiver, selected from 2,800+ applicants (2025)
- **IEEE Best Paper Awards** — 3× award recipient (2026)
- **Startup Bihar Grant** — INR 10L startup funding (2026)

Entrepreneurial Experience

Founder & CEO, Anu-Darshika AI Diagnostics *2025–Present*
AI-powered automated microscopy for medical/agricultural slide screening

- Developing a low-cost automated microscopy platform with a **12-slide scanning system** and deep-learning-based pathogen detection; supported by the **AICTE YUKTI Innovation Challenge** and NIELIT innovation funding.

Relevant Coursework

CS/AI	Machine Learning Techniques, Artificial Intelligence Techniques
Edge & Systems	Computer Organization, Operating Systems, Computer and Network Security
Math & Analytics	Statistics for Data Science, Numerical Methods for Data Science